UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 007 · Taylor's and Maclaurin's expansion · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

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1. Figure G1.1 · $\sin x$ and its Maclaurin polynomials
2. Figure G2.1 · The first terms of Taylor's series: tangent line and parabola
3. Figure G3.1 · $xe^y + 1$ and its first-degree Taylor polynomial
4. Figure G3.2 · Along the line $x = 1$: first- vs second-degree polynomial

Colours follow the Okabe–Ito order: blue = the function, vermilion = first approximation, green and purple for higher approximations; grey dashed lines are references. Figure G1.1 redraws the slide's graphs of $\sin x$ and its Maclaurin polynomials; the slide's own axis scale is not stated, so the redraw uses $-2\pi \le x \le 2\pi$. Figures G3.1–G3.2 redraw the deck's surface plots of $xe^y + 1$; the slides' screenshots are not copied.

1 Why Taylor's and Maclaurin's series?

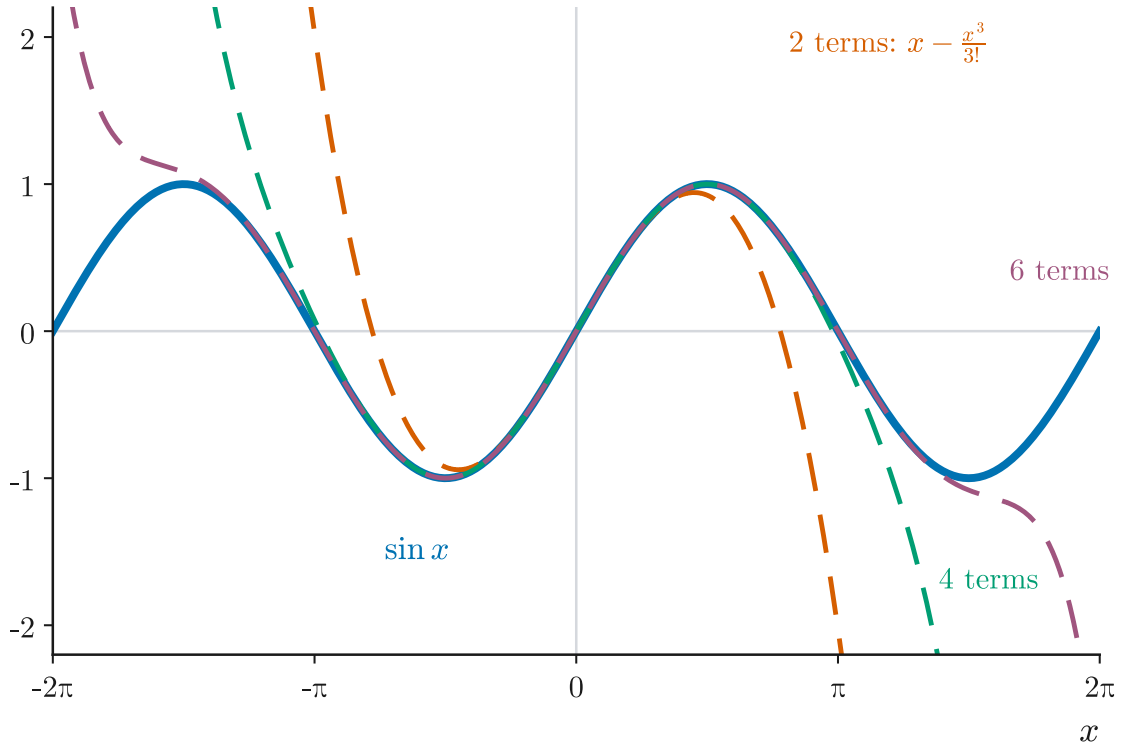


Figure G1.1. $\sin x$ and its Maclaurin polynomials. ESA ★

S007·8 · T007 00:14:45 · T007 00:15:58

How to read it. Blue is $\sin x$; the dashed curves keep 2, 4 and 6 terms of $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$. Each extra term extends the range where the polynomial follows the function: near 0 all of them agree; far from 0 the short polynomials shoot off. The slide shows 4, 6 and 10 terms; with 10 terms the curves overlap on this range.

2 Taylor's and Maclaurin's series in one variable

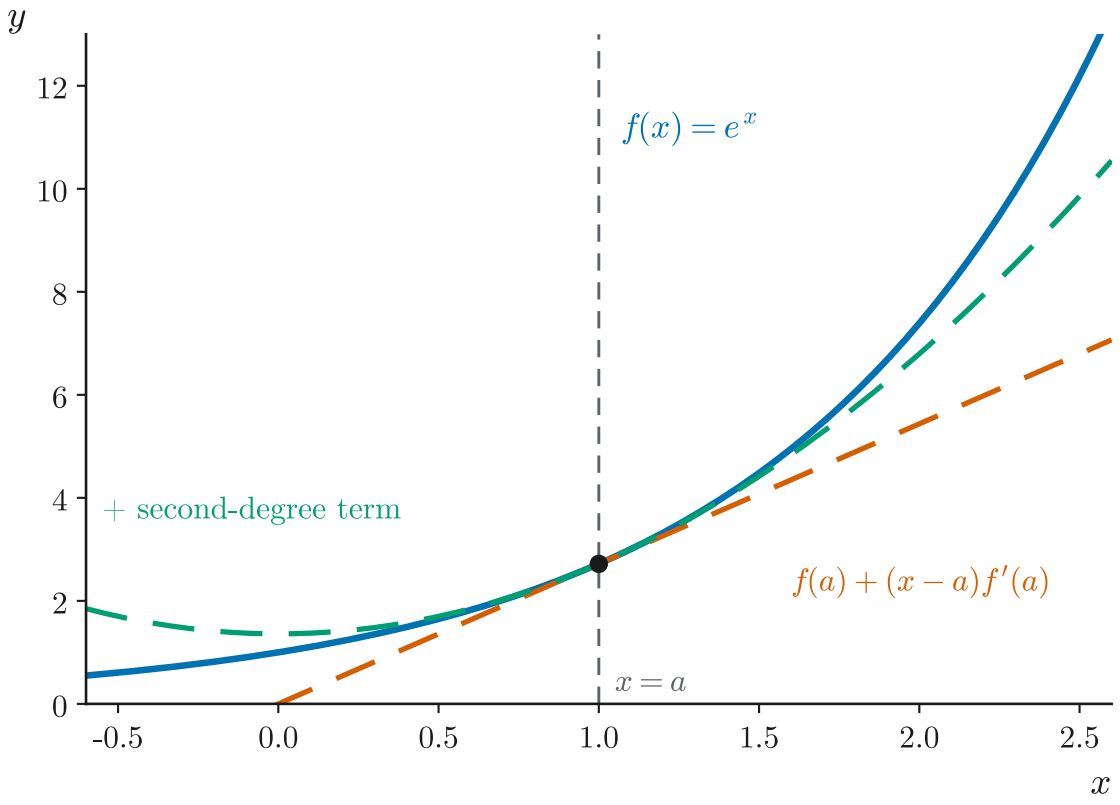


Figure G2.1. The first terms of Taylor's series: tangent line and parabola. ESA ★

ADDED FOR CLARITY T007 00:05:26 · T007 00:05:43

How to read it. $f(x) = e^x$ about $a = 1$. Two terms, $f(a) + (x-a)f'(a)$, give the tangent line (vermilion). Adding $\frac{(x-a)^2}{2!}f''(a)$ gives a parabola (green) that bends with the curve and stays close over a wider range.

3 Taylor's expansion of a function of two variables

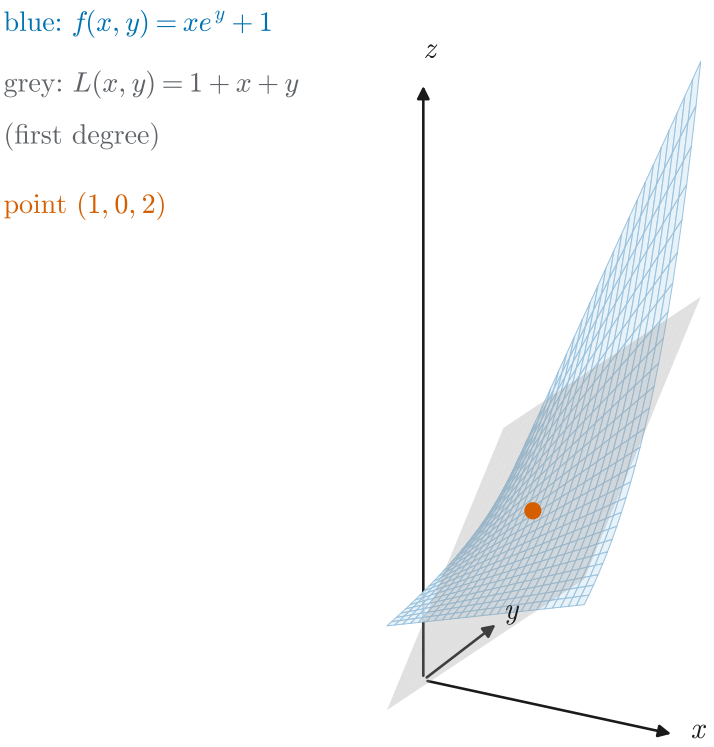


Figure G3.1. $xe^y + 1$ and its first-degree Taylor polynomial. ESA ★

S007·11 · T007 00:20:13

How to read it. The surface $z = xe^y + 1$ (blue mesh) and the plane $z = 1 + x + y$ (grey) touch at $(1, 0, 2)$. The first-degree Taylor polynomial is the tangent plane: it matches the surface's height and slopes at the point and separates from it away from the point.

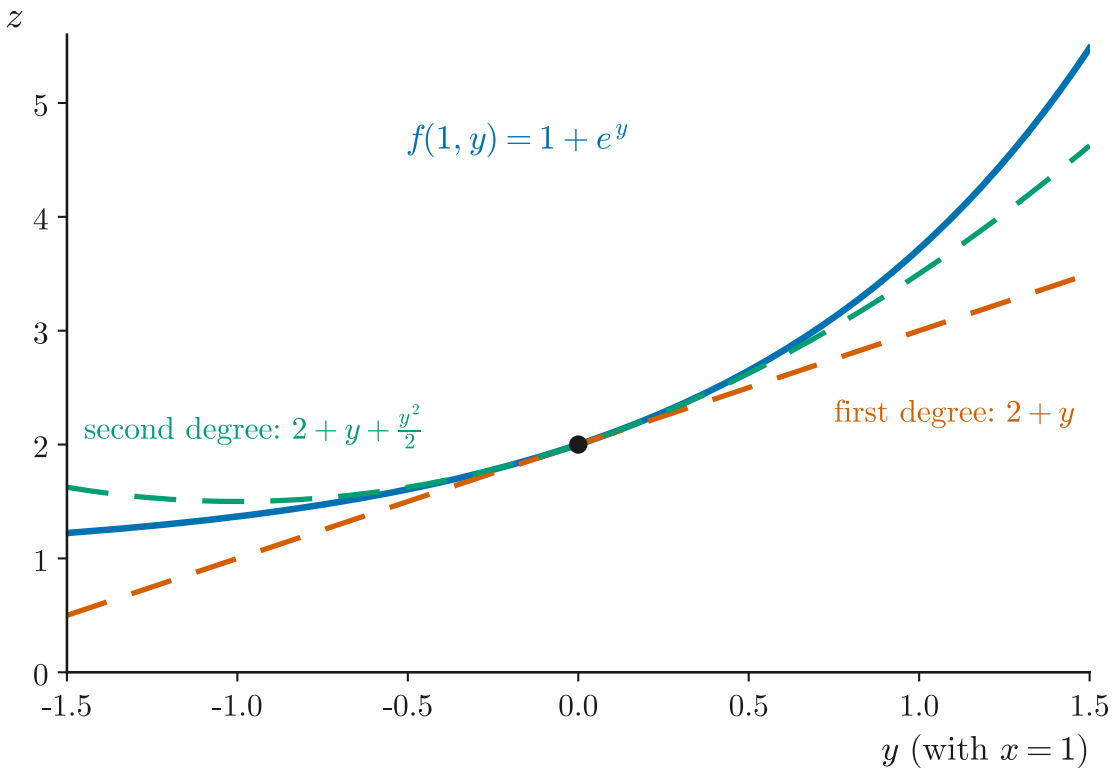


Figure G3.2. Along the line $x = 1$: first- vs second-degree polynomial. ESA ★

ADDED FOR CLARITY S007·11 · T007 00:20:51

How to read it. A slice of Figure G3.1 at $x = 1$: $f(1, y) = 1 + e^y$ (blue), the first-degree polynomial $2 + y$ (vermilion) and the second-degree $1 + x + xy + \frac{y^2}{2} = 2 + y + \frac{y^2}{2}$ (green). The second degree adds curvature and stays close much longer.